

Exp No: 8 Date:	Kuhn-Tucker conditions - Lagrangian Multiplier method
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AIM:

The objective of this lab is to introduce and implement the Kuhn-Tucker conditions for constrained optimization using Python. Participants will learn the theoretical basis of the Kuhn-Tucker conditions and apply them to solve optimization problems with inequality constraints.

PROCEDURE:

a. Objective Function and Constraints:

Define the objective function and inequality constraints in Python.

b. Kuhn-Tucker Conditions:

Implement the Kuhn-Tucker conditions using Python functions. Compute the gradients of the objective function and constraints.

c. Optimization Setup:

Use `scipy.optimize.minimize` or another suitable optimization function.

Incorporate the Kuhn-Tucker conditions as additional constraints or through custom callback functions.

d. Running the Optimization:

Execute the optimization procedure and monitor convergence. Validate results against theoretical expectations.

Problem-1:

Maximize: $Z = 2x_1^2 + x_2^2 + x_1x_2 + x_1 + x_2$

Subject to:

$$x_1^2 + x_2^2 \leq 1$$

$$x_1 \geq 0, x_2 \geq 0$$

Code:

```
import numpy as np
from scipy.optimize import minimize

# Objective function (maximize → minimize negative)
def objective(x):
    return -(2*x[0]**2 + x[1]**2 + x[0]*x[1] + x[0] + x[1])

# Constraint
def constraint(x):
    return 1 - (x[0]**2 + x[1]**2)

cons = {'type': 'ineq', 'fun': constraint} x0
= [0.5, 0.5]

result = minimize(objective, x0, constraints=cons)

print("Optimal Solution:", result.x)
print("Maximum Value:", -result.fun)
```

Output:

Optimal Solution: [0.79 0.61]
Maximum Value: 2.23

RESULT:

Thus, the Lagrangian Multiplier Method was successfully executed and verified using Python